

CAN EXPERIENTIAL GAMES AND IMPROVED RISK COVERAGE RAISE DEMAND FOR INDEX INSURANCE? EVIDENCE FROM KENYA

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Uninsured risk impedes agricultural production, but traditional indemnity insurance is not a viable option for smallholder farmers due to market failures. Weather index insurance is often touted as an alternative risk management tool. Demand for weather index insurance, however, has been weak. One possible reason is basis risk, the mismatch between index insurance payouts and individual outcomes. We estimate the impact of two interventions on preferred coverage under a real index insurance product, as revealed through an auction: (a) a reduction in basis risk, and (b) an experiential game that teaches farmers how index insurance works, with an emphasis on basis risk. We show all farmers demonstrate strong sensitivity to basis risk. The experiential game modestly increased knowledge, and experiential game participants indicate higher levels of preferred insurance coverage. Although offering a lower basis risk insurance product and playing an experiential game both increase preferred coverage in isolation, there is no additional impact of doing both. We adapt a theoretical model of misattribution bias to demonstrate how reference-dependent learning with loss aversion could lead to this unexpected result.

Key words: basis risk, games, index insurance, Kenya, misattribution bias.

JEL codes: D81, G22, O13, Q12, Q14, Q16.

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After decades of steady decline, progress toward halting world hunger has ceased. Eleven percent of today's global population is undernourished, and the number of people who suffer from hunger is slowly increasing (FAO 2019). According to one estimate the food supply will need to double by 2050 to satisfy increasing demand (Tilman et al. 2011). Uninsured risk creates a barrier to increased productivity by reducing incentives for investment, and weather risk in particular will likely increase in future years because of climate change (Dai 2012). Tools for reducing risk exposure, like agricultural insurance, could play an important role in raising agricultural productivity.

Available evidence suggests that access to agricultural insurance increases farm investment and thereby productivity in the U.S. (Cornaggia 2013) and developing countries alike (Mobarak and Rosenzweig 2012; Karlan et al. 2014; Cai et al. 2015; Cai 2016; Hill et al. 2019). But the effectiveness of standard indemnity crop insurance (as typically offered in developed countries) is undermined

by moral hazard (Babcock and Hennessy 1996; Smith and Goodwin 1996; Fadhliani, Luckstead, and Wailes 2018) and adverse selection (Just, Calvin, and Quiggin 1999). These asymmetric information problems, in addition to high transaction costs associated with offering policies to widely dispersed smallholder farmers, may render standard indemnity insurance unworkable in low-income countries.

In contrast to standard indemnity insurance, index insurance payouts are determined by an index beyond individual farmer control, for example, rainfall or average yields over a large area. Relying on an index eliminates adverse selection and moral hazard while also reducing transaction costs by removing the need for loss verification. Index insurance is now a fixture of the safety net for livestock producers in the U.S. (Goodrich, Yu, and Vandever 2019) and has also been applied in many contexts throughout the developing world (Miranda and Farrin 2012; Platteau, de Bock, and Gelade 2017). Unfortunately, demand for index insurance has been for the most part extremely low (Binswanger-Mkhize 2012; Platteau et al. 2017). For example, a recent review found that uptake of unsubsidized products across four index insurance evaluations ranged from 6% to 18%, with farmers who did buy insurance generally opting to cover a small portion of their land (J-PAL 2016).

One potential explanation for low index insurance demand is “basis risk”—the mismatch between insurance payouts and individual losses. Basis risk is a byproduct of basing payouts on an index rather than individual outcomes. Previous studies have shown that index insurance products marketed in India (Clarke et al. 2012), Kenya (Jensen, Barrett, and Mude 2016), and the U.S. (Yu et al. 2019) all carry substantial basis risk. Basis risk reduces demand for index insurance by increasing risk exposure and making it harder to understand the correlation between farm losses and insurance payouts. Many studies suggest that basis risk is perhaps the greatest barrier to index insurance adoption (Hazell and Hess 2010; Miranda and Farrin 2012; Carter et al. 2014; Clarke 2016). But offering products with low basis risk will only increase insurance uptake if farmers understand the connection between basis risk and the value of an index insurance product.

We assess (a) whether farmer preferred coverage levels for a real weather index insurance product respond to changes in basis risk, (b) whether an “experiential” game that simulates

farming and index insurance decisions over several seasons can improve knowledge about basis risk and index insurance more generally, and (c) whether and how the experiential game affects preferred coverage and sensitivity to basis risk. To accomplish these goals, we conduct a randomized control trial with smallholder grain producers in Kenya. All farmers in our study attend one of eighteen basic information sessions about index insurance. A randomization at the session level determines whether a given basic information session describes a high basis risk (HBR) weather index insurance product or a low basis risk (LBR) product. Following each basic information session, individual farmers are randomly assigned to participate in an experiential game simulating farming with the insurance type assigned to their session (HBR or LBR) or an unframed “placebo” game with payments, duration, and difficulty similar to those of the experiential game. After participating in the experiential game or the placebo game, farmers bid on their assigned insurance product—the product insures against drought during the upcoming farming season—in a multiple price list auction (Anderson et al. 2007).

We have four main findings. First, farmers who play the experiential game score 4.4% higher on a short index insurance knowledge quiz than farmers playing the placebo game. Quiz scores were high among farmers who only received basic information and played the placebo game, leaving little room for improvement, but our results suggest that the experiential game modestly increased knowledge. Second, these experiential game participants indicate higher levels of preferred insurance coverage. Third, even after receiving only basic information about index insurance, farmers demonstrate sensitivity to basis risk. Among farmers participating in the basic information session and placebo game, those assigned to the superior LBR product stated 37% higher preferred coverage levels than those assigned to the inferior HBR insurance product. Last, offering a lower basis risk insurance product and playing an experiential game both increase preferred coverage in isolation, but there is no additional impact of doing both. In other words, index insurance product quality and knowledge of index insurance appear to be substitutes rather than complements with respect to increasing preferred coverage levels.

We explore several possible reasons why farmers who play the experiential game appear less sensitive to basis risk than farmers

who do not and ultimately cannot rule out two possibilities. First, it could be that farmers have some maximum amount of insurance they will purchase at a given price, and that this maximum can be reached either by offering a better insurance product or by playing the experiential game. Second, it could be that farmers have reference-dependent preferences (Koszegi and Rabin 2006), and that reference points immediately after receiving the basic information session are based on overly pessimistic beliefs about the value of HBR insurance and excessively optimistic attitudes about LBR insurance. Our results would then be consistent with recent theoretical (Gagnon-Bartsch and Bushong 2019) and empirical (Bushong and Gagnon-Bartsch 2019) evidence that people incorporate feelings of disappointment or elation into economic valuations. We employ a theoretical model from Gagnon-Bartsch and Bushong (2019) to illustrate this.

Our paper makes three main contributions to the literature. First, we experimentally show that farmers can discern basis risk levels, even after a basic explanation, and prefer a product with reduced basis risk. Although several studies document the severity of basis risk carried by existing index insurance products (Clarke et al. 2012; Jensen et al. 2016; Yu et al. 2019), our paper is among only a few studies that test whether farmers are sensitive to variation in basis risk. Using longitudinal data, Jensen, Mude, and Barrett (2018) demonstrate lower insurance demand among Kenyan pastoralists with demonstrably high basis risk based on past shocks. Three separate studies from India also demonstrate that insurance demand decreases with the amount of basis risk. Using observational data, Hill, Hoddinott, and Kumar (2013) find that demand decreases with distance from the weather station used to determine insurance payouts. Mobarak and Rosenzweig (2012) randomize rainfall gauge location and find that demand decreases with basis risk, but that this relationship weakens where community risk sharing is strongest. Similarly, Hill, Robles, and Ceballos (2016) randomly introduce weather stations and find that insurance demand decreases with distance to a weather gauge. Unlike the aforementioned four studies, we experimentally alter the quality of the insurance product itself, rather than relying on variation in the amount of basis risk faced by individual farmers based on their location, which is potentially correlated with other factors affecting insurance demand.

Our findings on farmers' sensitivity to basis risk are relevant not only in Kenya and other developing countries where weather index insurance is being widely piloted but also in the U.S. where PRF-RI has existed for many years (Yu et al. 2019), and a weather index-based crop disaster program has been recently proposed as a far less expensive alternative to standard indemnity insurance (Belasco, Cooper, and Smith 2020). If index insurance is to be viable as more than a highly subsidized or niche product, our findings demonstrate the importance of improving quality by minimizing basis risk.

Second, we add to the literature on agricultural extension tools to teach farmers about complex products. In the case of index insurance, shocks that induce payouts are often infrequent and covariate; thus, learning from actual experience is a slow process with limited ability to observe and learn from others (Lybbert and Bell 2010). The evidence on the effectiveness of informational interventions on insurance demand, even intensive ones, is mixed. Gaurav, Cole, and Tobacman (2011) and Hill et al. (2016) show substantial increases in insurance uptake in India, whereas Cole et al. (2013) and Takahashi et al. (2016) show none in India and Ethiopia, respectively. Experiential games provide an interactive and engaging alternative to traditional informational interventions. Games have been used as an extension tool to explain insurance in other contexts, including projects in Kenya (McPeak, Chantarat, and Mude 2010), Ethiopia (Norton et al. 2014), Peru (Boucher and Mullally 2010), Mali (Elabed and Carter 2015), and Burkina Faso (Serfilippi, Carter, and Guirkingier 2018). To our knowledge, only Cai and Song (2017) and Jensen et al. (2018) have measured the impact of playing such games on insurance demand. Cai and Song (2017) find that playing insurance games in China led to large increases in indemnity insurance uptake, and that random shocks experienced within the game affect actual insurance demand. The game used by Cai and Song (2017) was designed to simulate standard indemnity insurance and therefore did not include any basis risk. Jensen et al. (2018) estimate observed basis risk for an index based livestock insurance product using eight seasons of panel data. They estimate the effect of a game (described in McPeak et al. (2010)) interacted with realized basis risk on insurance demand and find that playing the game increases knowledge of insurance as well as demand, particularly among farmers facing less basis risk. One concern with

their approach is that their observed indicator of basis risk is endogenous to the decision to insurance in previous years through changes in input use. In our study the amount of basis risk is randomly assigned, eliminating concerns about endogeneity.

We show experiential activities can improve learning and in some cases increase desired coverage amounts for some products. Our findings also suggest that the effectiveness of such approaches may be limited because of how new information interacts with prior perceptions of an insurance product and/or non-informational constraints on insurance demand. Although our paper focuses on using experiential learning to teach farmers about index insurance, experiential learning strategies could be developed to increase any agricultural technology plagued by low adoption levels globally where learning is a dynamic process (e.g. conservation agricultural practices).

Third, we apply a misattribution model to the case of learning about a technology. Several recent studies have examined the importance of reference points in the agricultural technology adoption literature (Liu 2013; Liu and Huang 2013; Magnan et al. 2020), including agricultural insurance (Babcock 2015; Feng, Du, and Hennessy 2019; Dalhaus, Barnett, and Finger 2020; Shin et al. 2020). These studies focus on risk preferences and employ reference levels of income or health. Here, we show how farmers' priors about the quality of a technology can serve as a reference point when learning about it, leading to misattribution bias in technology valuation (Bushong and Gagnon-Bartsch 2019; Gagnon-Bartsch and Bushong 2019).

The rest of this paper proceeds as follows. We begin by describing the sample, experimental design interventions, and the auction to elicit preferred insurance coverage levels. Next, we present our main empirical results before providing further exploration of our more surprising findings. We end with a brief conclusion.

Study Description

Study Setting and Recruitment

Our study takes place in Kenya, where agriculture plays a key role in economic growth, food security, and poverty reduction. In a recent World Bank report, D'Alessandro et al. (2015) state "Because the vast majority of Kenya's

poor depend on smallholder agriculture for their livelihood, increasing their productivity can contribute at once to improving food availability, increasing rural incomes, lowering poverty rates, and growing the economy." The same report presents evidence showing increasing frequency of severe droughts in Kenya as well as rising levels of year-to-year rainfall variability.¹ Combined with high dependence on rainfed agriculture and high poverty rates in rural areas, smallholder farmers in Kenya are particularly vulnerable to the effects of droughts. The same authors conclude that "erratic rainfall, punctuated by severe droughts, is the biggest risk facing Kenya's agriculture sector."

Drought risk is especially prevalent across the 80% of Kenya that is classified as arid and semi-arid lands, where approximately 30% of Kenyans reside. Farmers in our sample live in semi-arid Tharaka Nithi subcounty, located in the greater Meru region of the former Eastern Province, near Mt. Kenya. Most people in Tharaka Nithi are subsistence farmers growing sorghum or green gram, with a smaller number growing millet, maize, cow peas, or pigeon peas. Sorghum is more tolerant of hot and dry conditions than maize (Young and Long 2000), but drought stress at key times can have a strong negative effect on yield (de Camargo and Hubbard 1999). Similarly, green gram is more tolerant of drought stress than common bean (the most widely planted legume in the world) but still highly sensitive to the timing of water delivery (Bourgault and Smith 2010). Households generally supplement their agricultural income with handicrafts or small trade, but farming is the main source of income. Despite being semi-arid, few if any households use irrigation, which is fairly representative of the nation as a whole. Farmers in Tharaka Nithi also rarely purchase fertilizer, similar to what has been observed in Kenya and across Sub-Saharan Africa (Duflo, Kremer, and Robinson 2011).

The sample used in our study consists of 487 semisubsistence smallholder farmers in Tharaka Nithi subcounty, all with less than ten hectares of land. Of these, 390 had previously worked with an aggregator who supplies inputs on credit and then buys output at the end of the season, and ninety-seven intended to do so in the near future and were already

¹Kenya experienced an extreme rainfall event during two of every three years on average from 1980–2011 (D'Alessandro et al. 2015).

in contact with the aggregator.² Using a list of farmers provided by the aggregator, the aggregator's representative contacted the lead farmer of each farmer group. The representative requested that lead farmers invite other members of their groups to a specific session, with the stipulation that any invited farmer ought to be able to make decisions about insurance purchases on behalf of their household. Lead farmers then informed group members that they had been selected to participate in an activity about agriculture. Farmers were told the time and place of the activity, and that as a part of the activity they would be able to purchase insurance at a discounted price. Farmers were further told that if they wanted insurance, they would be asked to commit to purchasing the insurance during the activity, although payments would be made later. Specifically, they would be asked to pay for insurance at the beginning of the planting season, when they are accustomed to purchasing inputs.

For each session, we worked with the aggregator and a local extension agent to assemble groups of approximately thirty-two farmers from various villages at a central location. These groups were generally pre-existing farmer groups or combinations of groups. Gathering farmers in this manner is standard procedure for the aggregator. Information sessions took place approximately two days after the aggregator's representative initially contacted farmers. We conducted two sessions per day, one in the morning and one in the afternoon, and the groups of farmers assembled by the aggregator and extension agents were assigned to morning or afternoon based on farmer availability. At the meeting place, if a particular farmer was not initially present for the activity, the survey team visited or called them to encourage them to come.

Activities took place in July 2017, shortly before the sorghum harvest following the long rainy season. The insurance product being sold was to cover the short rainy season, with

planting to begin in October and harvest to take place in February–March 2018. Insurance is typically sold at planting time or just before, but the project funding timeline required activities to take place earlier. In addition, heavily contested general elections were held in Kenya on August 8, 2017 (Human Rights Watch 2017), resulting in a large gap between our main activities and the time of follow-up insurance sales in September–October.

Basic Information Session

Farmers in our study were offered insurance to protect sorghum production against drought risk. We also offered green gram insurance but only to farmers who indicated they would be planting green gram and not sorghum for the upcoming season. Although farmers working with the aggregator have had insurance in the past, most were unaware because the aggregator paid all premiums and all insurance payments went to the aggregator to cover losses from input loans. Farmers had never individually purchased insurance nor received insurance payments themselves. Although the aggregator purchases farmer output, the insurance offered as part of the study is unrelated to any contract farming agreement. Insurance used in our study is paid for by individual farmers before planting, and payouts go directly to the farmer when the index is triggered. Given the novelty and complexity of this product, it was deemed necessary to at least provide a basic explanation prior to offering insurance to farmers.

The basic insurance information session is purposefully simplistic. The promoter conducting the session focused on the farmers' transition from subsistence to commercial farming and explained to farmers that “because you are now producing sorghum and green gram as a business, you need to protect that business.” The promoter explained that with weather index insurance it is possible for farmers to not receive a payout even if they have a poor crop. The promoter did not explain the system of triggers that determine farmer payouts (e.g., a shortfall from expected precipitation during the flowering phase of the plant). This simple explanation follows the status quo in this setting, where the implementing partner insists that technical insurance contract details will confuse and intimidate farmers. The promoter did, however, explain the size of the index area (either a 5 × 5 km

²The sample consists mostly of farmers who work with the aggregator for three reasons: (a) our implementing partner, ACRE Africa, had their contact information and could assemble them through the aggregator; (b) the aggregator could vouch for ACRE's legitimacy as an insurance provider; and (c) they are fairly homogeneous in their production and could therefore be covered by the same insurance policies. Given that our sample consists of current and future aggregator clients, we consider our sample representative of the population of aggregator clients. In tables A8 and A9 of the online supplementary appendix, we show that our results are robust to excluding farmers who were not yet aggregator clients.

square or a 10×10 km square), because this is how the two insurance contracts offered as part of our study differ, as described in the subsection below.

All farmers receive basic information, so we cannot estimate the effect of receiving this information on preferred insurance coverage levels. We also cannot estimate the effect of being offered different insurance products in the absence of the basic information treatment, nor can we estimate the impact of playing the experiential game in the absence of basic information. Because any index insurance sales effort would include basic information about the product, we do not see this as problematic.

Treatments

The study uses a randomized control trial (RCT) with two treatments, insurance type and game type, that were assigned as follows. Each day, one session (morning or afternoon) was randomly assigned to HBR insurance while the other session was assigned to LBR insurance. Farmers and farmer group leaders had no way of knowing which insurance product would be offered in the morning or afternoon session when arranging which session to attend.

After a session was assigned to LBR or HBR insurance, individual farmers were randomly selected to either participate in an experiential insurance game or a placebo game that used a standard design to measure utility function parameters associated with prospect theory (Tanaka, Camerer, and Nguyen 2010). To determine game assignments, each farmer publicly drew a numbered card from an

envelope at the conclusion of his or her basic information session. Activities for the two game groups in each session (morning or afternoon) took place simultaneously and separately, with enough space between groups to avoid spillovers during session activities. Groups of between one and three farmers worked with a single enumerator to complete the steps of the assigned game. Participants offered LBR insurance played an LBR version of the experiential game and participants offered HBR insurance played an HBR version of the experiential game. A given farmer would only be exposed to one of the two insurance products regardless of their game treatment assignment. We discuss the details of the two insurance contracts and game types in the next subsection. Our assignment mechanisms yielded a 2×2 factorial design with four possible treatment combinations, summarized by the schematic presented in figure 1.

Table 1 presents summary statistics for the sample as a whole and disaggregated by treatment group. Appendix table A1 shows differences in means across the four treatment groups. When we examine the balance resulting from our randomization, seven out of sixty-six differences in means are statistically significant at the 10% significance level, slightly more than expected by chance. To mitigate the possibility of imbalance influencing our results, we control for all of the characteristics shown in table 1 in our regression specifications. We describe the different treatments in further detail in the subsections below.

Insurance treatment arms. The only available insurance product in the area prior to the

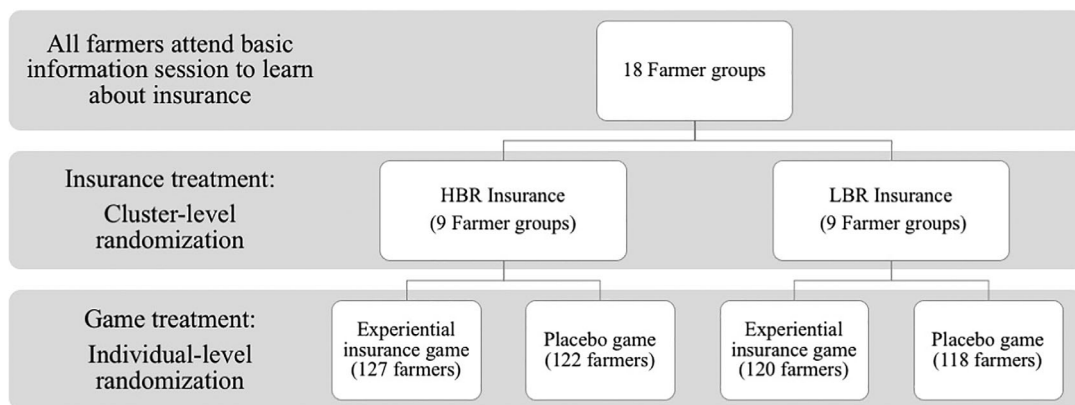


Figure 1. Illustration of study design

Table 1. Summary Statistics

Variable	Total	High basis risk insurance		Low basis risk insurance	
	Sample	Placebo game	Insurance game	Placebo game	Insurance game
Age (years)	40.372 (1.033)	40.139 (1.457)	40.614 (1.792)	41.119 (1.883)	39.617 (1.379)
Education (years)	6.947 (0.224)	6.934 (0.239)	6.323 (0.421)	7.508 (0.430)	7.067 (0.465)
Literacy (0/1)	0.743 (0.029)	0.705 (0.042)	0.685 (0.057)	0.780 (0.049)	0.808 (0.040)
Male (0/1)	0.308 (0.039)	0.270 (0.036)	0.346 (0.044)	0.254 (0.060)	0.358 (0.093)
Land size (acres)	4.309 (0.202)	4.110 (0.227)	4.490 (0.381)	4.854 (0.487)	3.783 (0.351)
Droughts last five years (count 1–5)	2.850 (0.055)	2.787 (0.081)	3.055 (0.101)	2.822 (0.131)	2.725 (0.127)
Can access loan (0/1)	0.665 (0.022)	0.664 (0.049)	0.638 (0.034)	0.695 (0.044)	0.667 (0.052)
Formal savings (0/1)	0.674 (0.024)	0.639 (0.045)	0.646 (0.060)	0.712 (0.038)	0.700 (0.059)
Planted acreage	1.951 (0.105)	1.863 (0.135)	1.999 (0.255)	1.970 (0.163)	1.971 (0.174)
Projected net crop income (KSH)	52,182.531 (5,317.441)	49,852.725 (9,903.871)	53,308.976 (13,479.180)	52,986.992 (10,645.034)	52,567.958 (4,941.703)
Green gram (0/1)	0.066 (0.015)	0.049 (0.026)	0.071 (0.021)	0.051 (0.026)	0.092 (0.032)
Observations	487	122	127	118	120
Game sessions	18	9	9	9	9

Note. The summary statistics shown are means and standard errors of the mean, with the latter in brackets. Insurance type was assigned at the game session level, whereas game type was assigned individually.

study was developed by ACRE Africa (ACRE, hereafter) using African Rainfall Climatology version 2 (ARC2) data. ARC2 data are generated using infrared cloud-top temperature measurements combined with 24 hour rainfall gauge observations from across Africa and are produced at 10 × 10 km resolution (Novella and Thiaw 2013). This means a single rainfall measure is calculated for each 10 × 10 km grid square. As an alternative, Climate Hazards Group InfraRed Precipitation with Station (CHIRPS) data are generated using infrared Cold Cloud Duration observations to estimate rainfall and produce estimates at a more refined 5 × 5 km resolution (Funk et al. 2015). In theory smaller grid squares means lower basis risk, but this may not be the case because of differences in how rainfall is estimated in the ARC2 versus the CHIRPS data. Our goal was to design two insurance policies that only differ in basis risk (i.e., the actuarially fair price of the two products is the same), so for the HBR insurance offered in the study we created rainfall estimates for 10 × 10 km squares using CHIRPS data (see part B of the online supplementary appendix for details).

Although we conducted the experiential game and the auction using both insurance products, farmers were later only given the opportunity to purchase LBR insurance for ethical reasons; we saw no reason to sell an artificially generated inferior product when a better one could be provided at the same cost. At the time of the auction, HBR participants were not aware of the existence of LBR insurance or that it would ultimately be made available to them. Instead, they were offered a free upgrade when they were later visited to complete final insurance transactions agreed to during the auction. Thus, the upgrade could in no way affect the auction data used in our analysis.

Experiential game treatment arms. The weather index insurance experiential game uses maps specifically designed to teach farmers about basis risk and also convey information on the frequency of negative weather shocks and the magnitude of insurance payouts. The maps are carefully designed to resemble simplified versions of the real insurance products offered to farmers. Maps consist of points (“farms”) and grid squares (index areas). Sample maps similar to those used in the experiential game are found in figures 2

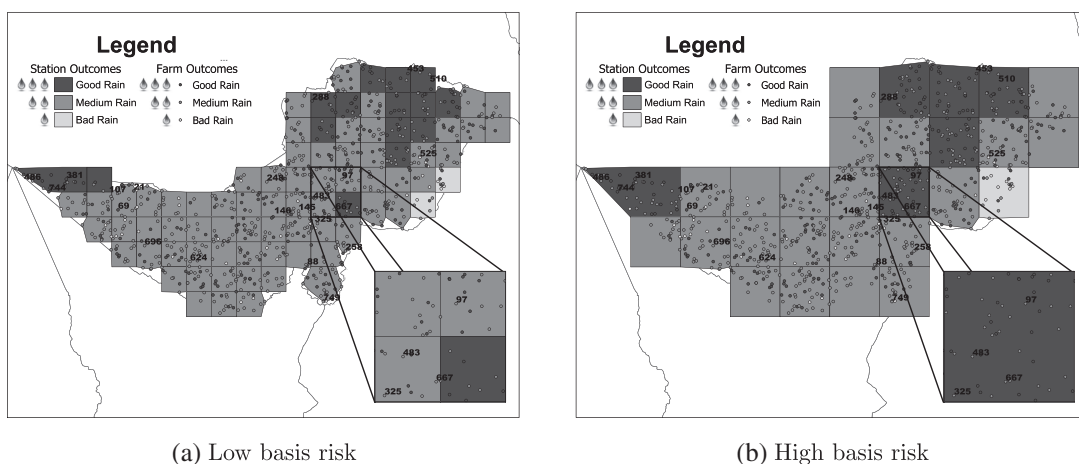


Figure 2. Farm and index area outcome maps

(a),(b) (for LBR insurance) and (for HBR insurance).³ Before the start of the experiential game, one of the farms is randomly assigned to each participant and used to calculate that farmer's payouts in each round. Production values for individual farms are determined at the farm level, whereas insurance payouts are determined at the square level, which reflects the main distinguishing feature of index insurance. There are three levels of rainfall depicted in the maps: "bad" (light gray), "medium" (medium gray), and "good" (dark gray).⁴ Farms and squares each take on one of these colors, with farms not necessarily being the same color as the square containing them. Mismatches between farm and grid square, which are evident in the inset shown in figure 2(a), are cases of basis risk events. A positive basis risk event is where the rainfall on the farm is better than on the associated square, and a negative basis risk event is where the rainfall on the farm is worse than on the associated square. The experiential game for HBR and LBR differs only in the size of their grid squares, just as the real insurance policies offered differ only in the size of their grid squares.

Using these maps, farmers experience nine simulated agricultural seasons, with two practice rounds followed by seven rounds with

stakes. We created a unique map for each season using actual historical data.⁵ The maps were presented in the same order each time the experiential game was played. The farms and nine years of the experiential game were chosen so that the probability of a negative basis risk event is substantially lower in the LBR insurance experiential game than in the HBR version, and the probability of positive basis risk events is roughly equal between the two. Figure 3 shows that our calibration of probabilities was successful. The overall probability of experiencing a negative basis event was about 12 percentage points higher with HBR than LBR, at 29% versus 17%. Put differently, the probability of a negative basis risk event was 73% higher with HBR than LBR. The difference in the round-by-round probability of a negative basis risk event ranges from nearly 1 percentage point in round 5 to 21 percentage points in round 1. For positive basis risk events, the difference in overall probabilities is less than a percentage point, with round-by-round differences ranging from near zero to 5.7 percentage points. Regression results in table A4 of the online supplementary appendix show that farmers assigned to LBR insurance who played the insurance experiential game experienced an average of 0.84 fewer basis risk events than farmers assigned HBR insurance, a reduction of 42% ($p < 0.01$), while there was no effect of contract type on the average number of positive basis risk events.

³Original experiential game maps were printed in color and did not show the insets.

⁴In reality, we used maps with three colors (brown, yellow, and green) but use black and white maps here for those reading printed copies of this document. The original maps are shown in figures B1a and B1b of the online supplementary appendix.

⁵The chosen years are 1988, 1990, 1991, 1995, 2002, 2004, 2007, and 2011. The maps pictured here use the 1991 data.

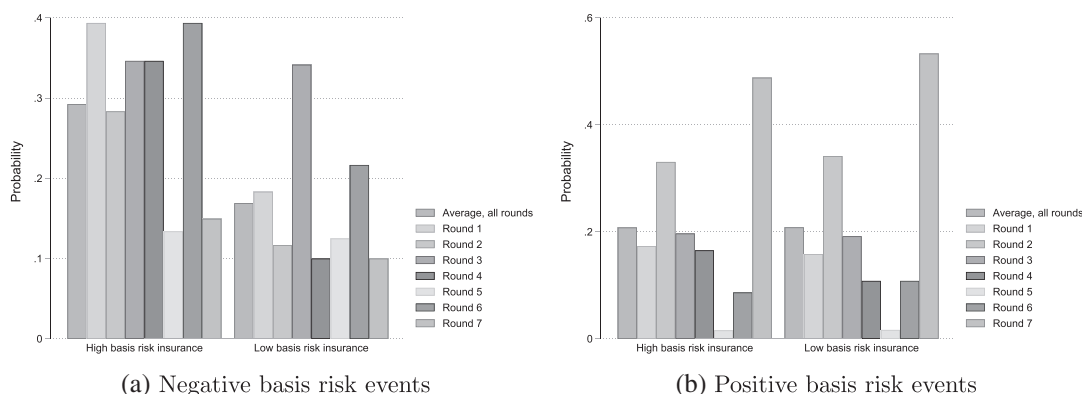


Figure 3. In-game basis risk event probabilities

At the beginning of each experiential game season, a random insurance price is selected from the following values: 100, 200, 500, 700, 1,000, 1,500, and 2,000 game shillings, where one Kenyan Shilling (KSH) = 100 game shillings.⁶ Farmers are then asked whether they would like to purchase zero, one, or two units of insurance at the drawn price. After each farmer decides how much insurance to purchase, a map displays farm-level outcomes (not including insurance payouts). Bad years, medium years, and good years (at the farm level) pay out 5,000, 7,000, and 10,000 game shillings, respectively. Once all farm-level outcomes are revealed, a second map depicting both farm- and square-level results is revealed. One unit of insurance pays out 2500 game shillings in a bad year (at the square level), 1,500 in a medium year, and nothing in a good year. These values were chosen so that two units of insurance fully compensate a farmer for the corresponding level of loss.

To engage farmers and make the experience of gains and losses in the experiential game more salient, we gave farmers real payments according to their performance. To avoid dynamic play, we told farmers at the beginning of the experiential game that payouts would be based on a single randomly selected round, excluding the two practice rounds. To mitigate income effects, we calibrated average game payouts in the experiential game and the placebo game to be equal.⁷ Furthermore, the

amount of money given to farmers is very small relative to the actual price of insurance. Every farmer playing the experiential game was given a minimum payment of 100 KSH. On top of that, farmers could earn an additional 10 to 149 KSH from the experiential game.⁸ Including the minimum 100 Ksh payment, average experiential game payouts were 199 Ksh for LBR and 196 Ksh for HBR. At actuarially fair prices, insuring one acre of sorghum production (outside of the experiential game) costs approximately 3,600 KSH. Thus, we do not believe the income effect would result in differences in auction bids or eventual insurance purchases. Further details of experiential game design can be found in part B of the online supplementary appendix. The protocol for the experiential game can be found in part C of the online supplementary appendix.

There are several mechanisms by which the experiential game could influence behavior in the insurance auction: (a) farmers learn about index insurance and basis risk; (b) farmers become tired from playing the experiential game, which lasts approximately one hour and consists of many decisions; and (c) farmers interact at length with the enumerators, who also conduct the auction; and (d) farmers receive a payout that is based on both experiential game performance and luck. In order to isolate (a), we administered a placebo game to farmers not playing the experiential game (treatments 2 and 4 listed above) to net out

⁶100 KSH ≈ \$1 US.

⁷By design, the average experiential game payouts and the probability of paying out are the same for both insurance products, although farmers assigned to LBR insurance earned slightly more on average because they experienced fewer negative basis risk events.

⁸A farmer would earn an additional 149 KSH by purchasing two units of insurance at 100 game shillings each and then experiencing good rain at the farm level and bad rain at the square level. A farmer would earn an additional 10 KSH by purchasing two units of insurance at 2,000 game shillings each and then experiencing bad rain at the farm level and good rain at the square level.

(b), (c), and (d). Specifically, we use prospect theory preference elicitation games (Tanaka et al. 2010). Like the experiential game, the placebo game takes approximately one hour to complete and consists of many decisions. Payouts are also dependent on both game play and luck. The placebo game should not net out (a) as the game has no explicit mention of weather, farming, or insurance, nor does it have any basis risk. Although it is possible that playing the placebo game would prime respondents to think about risk in some way, perhaps by making risk more salient, we would still manage to isolate all learning about index insurance and basis risk, which is what we are ultimately interested in. The protocol for the placebo game can be found in part C of the online supplementary appendix.

Auction

Immediately after participants played their assigned game, we conducted a multiple price list auction to capture preferred insurance coverage levels. In India, Cole et al. (2016) show that a binding auction for a fixed amount of weather index insurance coverage results in uptake rates similar to those from an auction in which each farmer faces a single randomly assigned price. We allow producers to choose their desired coverage level at several prices before drawing a single price at which purchase commitments occur, using a method similar to Lybbert et al. (2013).

Insurance quantities are coverage amounts (in KSH), whereas premiums (prices) are measured as a percentage of the amount covered. During piloting, we found that farmers were confused by insurance quantities expressed as monetary amounts (that could take on any value) and prices expressed as percentages. After piloting different ways to auction weather index insurance, we opted to sell “units” of insurance, where each unit provides 5,000 KSH of coverage. We express prices for each unit of insurance in amounts (50, 150, 250, 350, 500, 750, and 1,000 KSH) that correspond to percentages (1, 3, 5, 6, 10, 15, and 20%) of 5,000 KSH of coverage. At each price, farmers state the number of units they wish to purchase.

The auction unfolds as an iterative process in which an enumerator first tells a farmer how many units of insurance it would take to insure her entire crop value, and how much this would cost, at a very low price (e.g., 50

KSH). The farmer typically agrees to insure the entire value of her crop at that price. The enumerator then proceeds to raise the price of a unit of insurance to successively higher levels, and the farmer states a desired number of insurance units at each new price. Once all prices in the list have been covered, or a price is reached at which the farmer indicates zero coverage, a single price is randomly selected as the actual price. The enumerator then confirms with the farmer that they are committing to purchase the amount of coverage they said they wanted at that price. The auction protocol can be found in E of the online supplementary appendix.

Two-way scatterplots of the stated preferred coverage level at different prices are shown in figure 4. We winsorize the insurance coverage data at the 99th percentile (150,000 KSH) to reduce the influence of outliers in our analysis. These winsorized observations can be seen at the top of the subfigures. Each subfigure displays stated preferred coverage under a given treatment arm. The vertical axis shows prices and the horizontal axis shows stated value insured at each price. Prices are constrained to the seven prices offered and coverage levels are constrained by units purchased (each unit provides 5,000 KSH of coverage), so we use the “jitter” function in Stata to add random noise along both axes. This allows us to see more data points, although we can still see the bins for the various prices offered.⁹

The figure provides a descriptive preview into our results. Across the four treatments, 2,473 of 3,409 values of preferred coverage levels are non-zero. Figure 4(a) shows the preferred level of coverage at various prices for all individuals assigned the placebo game and offered HBR insurance. Figure 4(b) shows the preferred coverage at various prices for individuals assigned the experiential game and offered HBR insurance. When moving from figure 4(a) to 4(b), it's evident that the preferred coverage with HBR insurance is higher with the experiential game. Figure 4(c) shows preferred coverage for those assigned to play the placebo game and offered LBR insurance. Comparing figure 4(a)–(c) reveals that among those assigned to the placebo game, we observe higher preferred coverage for the lower basis risk product.

⁹The jittering results in the winsorized observations not all being at exactly 150,000 but at values near 150,000.

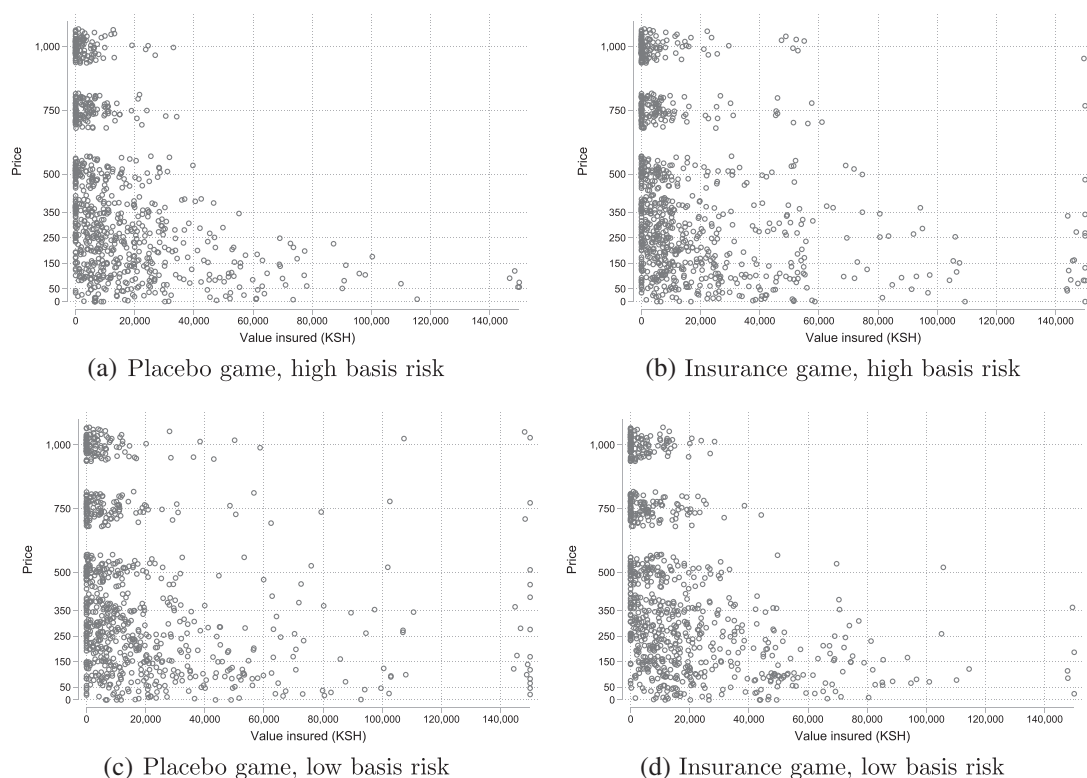


Figure 4. Preferred stated coverage level at different prices by treatment type

Finally, figure 4(d) shows the preferred coverage for those assigned to play the experiential game and offered LBR insurance. This figure provides a first glimpse of our finding that although the insurance game and LBR insurance both increase preferred coverage across a variety of prices, combining the interventions does not seem to have an additional positive effect.

The enumeration team stressed throughout the auction that farmers were expected to pay for coverage as determined by their bids. In the end, however, the auction was not binding. There are a number of reasons why we could not collect payment immediately following each session's activities. First, our implementation partners emphasized we could not expect farmers to make such a large investment without providing time to save. Second, the funding timeline and the agricultural season were misaligned—funding requirements for our research grant required spending all resources just before the time when insurance would normally be sold. Third, expectations for turbulent elections required all field work to be completed months before farmers would normally be making agricultural investment

purchases for the season of interest. These features collectively led to a large time gap (approximately two months) between collection of the stated preferred coverage data and the actual sales. Had it been possible to follow-up more quickly, the realized demand may have more closely reflected stated preferred coverage levels. Unfortunately, we have no way of knowing what revealed demand would have been had we required purchases on site or how requiring farmers to purchase insurance on site would have changed stated coverage levels in the auction. We discuss insurance purchases in greater detail in part F of the online supplementary appendix.

Our analysis thus relies on preferred stated coverage value from the auction not realized insurance demand. A large literature suggests stated willingness to pay or contingent valuation survey data are unlikely to correspond exactly with revealed preference data obtained via actual purchase decisions. Some have argued that all contingent valuation surveys are misleading and their use is misguided (Diamond and Hausman 1994). In a similar context and application to our own study, McIntosh, Sarris, and Papadopoulos (2013)

find that stated willingness to pay and actual demand for index insurance in Ethiopia are very poorly correlated. Contrarily, in social psychology the theory of planned behavior implies stated intentions do indeed result in realized behavior (Ajzen 1991). Meta-analyses in Albarracín et al. (2001), Sheeran (2002), and Webb and Sheeran (2006) show strong empirical support for the notion that stated intentions predict actual behavior.

We acknowledge the limitations of using stated coverage levels from the auction and do not claim our stated coverage data are necessarily reflective of realized demand. Despite the limitations of stated coverage data, we argue it is still highly relevant in this case. Even if stated coverage is not equivalent to true demand, it can still be a signal of the value placed on something. Any observed differences between stated coverage levels across treatments ought to reflect differences in preferences for insurance that stem from the treatments. These different preferences are revealed in the auction, even if they are not reflected through realized purchases of insurance.

Empirical Approach and Results

In this section, we first estimate average treatment effects of playing the experiential game on learning outcomes. We then employ the auction data to test the effects of being offered LBR insurance, playing the experiential game, and the combination thereof, on stated preferred weather index insurance coverage. This study has a pre-analysis plan registered with the American Economic Association.¹⁰ Our learning analysis deviates from the pre-analysis plan, whereas the auction data analysis closely follows the plan. All deviations from the pre-analysis plan and the rationale behind them are described in part G of the online supplementary appendix.

Learning

If the experiential game is an effective learning tool, we would expect playing it to make farmers more knowledgeable about weather index insurance. At the conclusion of the auction we asked farmers four true/false questions to test their knowledge about how weather

index insurance works. The questions are as follows, with the correct answers in brackets:

1. If I buy rainfall insurance, I will always receive money back at the end of the season, regardless of the weather. [False]
2. It is possible to receive an insurance payout even if I have received enough rain on my fields. [True]
3. To determine how much rain has fallen, the insurer measures average rainfall over larger squared areas, not at a single farm. [True]
4. The smaller the square area used by the insurer to measure rainfall, the greater the similarity between rainfall measured by the insurer and rainfall on my fields. [True]

Note that a better understanding of these points does not necessarily translate to higher or lower demand for index insurance. For example, if a farmer learns that they will not always receive a payout, their demand could decrease. If they learn they might receive a payout even if they receive adequate rainfall, their demand might increase or decrease depending on whether they also understand they might not get a payout if they do not receive adequate rainfall. If they learn that rainfall is determined at an aggregate level rather than at the farm level, their demand would likely decrease. If they learn that a smaller grid square results in aggregate rainfall measurement that is more similar to the rainfall on their farm, their demand would likely increase or decrease based on how large they interpret the squares to be. We therefore cannot draw a straight line from learning about weather index insurance to an increase or decrease in preferred coverage levels.

We add the number of correct answers together to generate a knowledge score and estimate the effect of playing the experiential game on this score as well as on whether they answered each question correctly. We do not differentiate here between playing the experiential game with the HBR or LBR product because each version of the experiential game should convey the same information (in the discussion following our empirical results we address the possibility that they do not). The estimation model is:

$$(1) \quad k_i = \psi_0 + \psi_1 G_i + \mathbf{X}_i' \boldsymbol{\psi}_x + e_i$$

where i indexes farmer, k_i is the knowledge index (0–4) or a binary variable for answering

¹⁰The pre-analysis plan can be found at <https://www.socialscisearch.org/trials/2401/history/21197>

an individual question correctly, G_i indicates treatment, and $\mathbf{X}_i$ contains the variables from table 1, which are included as controls to improve precision. When estimating treatment effects on the knowledge score we use ordinary least squares (OLS), and when estimating treatment effects on binary indicators we use logistic regression and report marginal effects. Our results are robust to using linear probability models rather than logistic regression.

Learning results can be found in table 2 below. On average, farmers who do not play the experiential game but do participate in the basic information session answer three of four questions correctly. The high scores in the control group suggest that the basic information was effective given that farmers reported having very little prior knowledge of insurance, although we cannot test this empirically.

We find that playing the experiential game increases the knowledge score by 0.13, or from the average of 2.967 correct answers in the sample not playing the experiential game to an average of 3.097 correct answers, an increase of 3.25 percentage points or 4.4%. For individual questions, without the experiential game 82% of farmers understood that they would not always receive a payout from insurance, 46% of farmers understood they might receive a payout even if they had adequate rain on their farm, 93% understood rainfall was measured over a larger squared area rather than at the farm level, and 75% of farmers understood that a smaller square would result in rainfall measurement more similar to the rainfall on their farm, as opposed to a larger square. The experiential game has a significantly positive effect on answering the third (5 percentage points) and fourth (7.8 percentage points) questions correctly. It has a positive but insignificant effect on answering the first question correctly, and a negative and insignificant effect on answering the second question correctly. This negative coefficient is unsurprising, as the majority of farmers playing the experiential game would not experience a positive basis risk event.

Based on these results we conclude that the experiential game was effective at increasing knowledge about insurance, but that knowledge was fairly high after the basic information session. It is important to keep in mind that the experiential game is designed to teach farmers not only about the basics of weather index insurance but also about the frequency of

negative weather shocks, the magnitude of payouts, and the frequency of basis risk events. Playing the experiential game is expected to influence preferred coverage through these channels, and the degree to which it does may depend on whether farmers play the experiential game that corresponds to the HBR or LBR insurance product. In the following section we investigate the effect of being offered a LBR product (as opposed to a HBR product), playing the experiential game that corresponds to that product, and the combination thereof on preferred insurance coverage levels.

Insurance Coverage

To estimate the impacts of the different treatments on preferred insurance coverage levels, we regress stated preferred coverage data at all prices on treatment status, controlling for price and a list of covariates. We have seven observations per farmer, one at each price, for a total sample size of 3,409. The model is specified as:

$$(2) \quad Q_{ij} = \beta_0 + \beta_1 G_i + \beta_2 LBR_i + \beta_3 G_i \times LBR_i + \beta_4 P_j + \mathbf{X}_i' \boldsymbol{\beta}_x + \varepsilon_{ij}$$

In equation (2), Q_{ij} is the total value (in KSH) to be insured by farmer i at price j as stated in the auction. G_i and LBR_i are indicators for random assignment to the experiential game and LBR insurance, respectively. The omitted treatment category includes farmers who did not play the experiential game and were offered HBR insurance, which we refer to as the status quo treatment. P_j is the price at which the preferred insurance coverage was reported, expressed as percent of coverage, and $\mathbf{X}_i$ contains control variables shown in table 1.

The average treatment effect of LBR insurance for farmers assigned to the placebo game is β_2 , while the average treatment effect for farmers playing the experiential game is $\beta_2 + \beta_3$. The overall average treatment effect of LBR insurance is $\beta_2 + \beta_3 \pi_G$, where π_G is the probability of being assigned to the experiential game (set to the observed proportion of the sample assigned to the experiential game). We expect β_2 to be positive but smaller than $\beta_2 + \beta_3$, given that the experiential game increased knowledge (although only modestly). The overall average treatment effect of LBR should therefore be positive.

Table 2. Treatment Effects on Knowledge

Outcome	Game treatment effect	No game	Observations
Knowledge score (1–4)	0.130* (−0.009, 0.270)	2.967 (0.050)	487
Insurance does not always pay out (0/1)	0.051 (−0.015, 0.117)	0.817 (0.025)	487
Insurance may pay out even when I have enough rain (0/1)	−0.052 (−0.140, 0.036)	0.463 (0.032)	487
Rainfall is measured over an area, not a single farm (0/1)	0.044** (0.004, 0.084)	0.933 (0.016)	487
Smaller area = better correlation with my rainfall (0/1)	0.087** (0.017, 0.157)	0.754 (0.028)	487

Note: First row estimated using OLS, rows 2–5 estimated using logit (marginal effects shown). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. 95% confidence intervals based on heteroscedasticity-robust standard errors in brackets. The “No game” column shows averages of each indicator among farmers not playing the experiential game, with the standard error of the mean in parentheses. Covariates include age, education, literacy, gender, land size, number of droughts in the past five years, loan availability, formal savings, planted acreage, projected net value of production, and crop (green gram or sorghum).

The average treatment effects of the experiential game for farmers assigned to HBR insurance and LBR insurance are β_1 and $\beta_1 + \beta_3$, respectively. The overall average treatment effect of the experiential game is given by $\beta_1 + \beta_3\pi_{LBR}$, where π_{LBR} is the proportion of farmers assigned to LBR insurance. The expected signs of the experiential game effects are ambiguous for reasons discussed earlier: simulating index insurance could raise preferred coverage levels by demonstrating its value as a risk management tool or decrease preferred coverage levels by showing the damage done by basis risk. That being said, because we expect a positive interaction effect between playing the experiential game and being offered LBR insurance (i.e. $\beta_3 > 0$), the impact of the experiential game on preferred coverage levels ought to be greater for farmers offered LBR insurance than farmers offered HBR insurance.

In addition, we re-estimate equation 2 twice: once while limiting the sample to preferred coverage levels at the actuarially fair price (350 KSH per 5,000 KSH of coverage) and again at the market price that would be normally charged according to the insurer (750 KSH per 5,000 KSH of coverage). Because we condition on a single price, we drop P_j from our set of controls, but otherwise our regression specification is identical to equation 2, although with one observation per farmer rather than seven. Our motivation for examining impacts at specific prices is that many of the prices included in the auction were very low relative to those that farmers might actually encounter.

Point estimates and 95% confidence intervals for treatment effects on preferred coverage levels estimated using all prices as well as the actuarially fair price and the market price

are found in table 3. Full regression results for the model with all prices can be found in table A2 of the online supplementary appendix. When performing inference we cluster at the level of treatment assignment (Abadie et al. 2017). Because game type was assigned individually while insurance products were assigned by session, we cluster by farmer for experiential game effects and cluster by session for insurance effects. We use the wild cluster bootstrap for insurance product effects because there were eighteen total sessions, which is too few for analytic cluster standard errors to be reliable (Webb 2013; Cameron and Miller 2015).

The top row of the first column in table 3 shows average stated preferred insurance coverage for farmers in the status quo treatment group, that is, offered HBR insurance and not playing the experiential game. Among farmers in the status quo treatment, the average preferred coverage levels are KSH 13,460, KSH 10,963, and KSH 3,955, respectively, at all prices, the actuarially fair price, and the market price. The second row contains the average experiential game effect on preferred insurance coverage levels, calculated as $\beta_1 + 0.486\beta_3$, where 48.6% of farmers in the experiential game treatment group were also in the LBR treatment group. When looking at the results for all prices, those playing the experiential game have preferred coverage levels that are 1,292 KSH (9.6%) higher than the status quo group, although this difference is not statistically significant.

The third row contains the experiential game effect for those offered HBR insurance, β_1 , and the fourth row contains the experiential game effect for those offered LBR insurance, $\beta_1 + \beta_3$. Playing the experiential game increases preferred coverage levels by 3,864

KSH (29%) among farmers assigned to HBR insurance, and the effect is significant at the 5% level. For those offered LBR insurance there was no statistically significant effect of the experiential game, and the point estimate is -1,430.

The fifth row of table 3 contains the average difference in preferred insurance coverage levels for LBR insurance compared to HBR insurance, calculated as $\beta_2 + 0.504\beta_3$, where 50.4% of farmers in the LBR treatment group also played the experiential game. The sixth row contains the estimated effect of LBR insurance on preferred insurance coverage levels for those who did not play the experiential game, β_2 ; and the seventh row contains the estimated LBR effect for those who did, $\beta_2 + \beta_3$. As expected given the results described in the previous paragraph, we find higher preferred insurance coverage levels for LBR insurance only among farmers who did not play the experiential game. Among those who did not play the experiential game, preferred coverage levels for LBR insurance is 4,961 KSH (37%) higher than preferred coverage levels for HBR insurance. Preferred coverage levels for LBR and HBR insurance is nearly identical among those who did play the experiential game (the point estimate of the difference is -332 KSH).

Results at the actuarially fair and market prices are shown in the second and third columns of table 3, respectively. On average we find no significant effect of playing the experiential game at either the actuarially fair or market price. The point estimate at the actuarially fair price is larger (2,229 KSH) than we find when using all prices (1,292 KSH), whereas the point estimate at the market price is close to zero (17.2). When we estimate the experiential game effect separately for those offered HBR and LBR insurance, we find a significant increase of 5,374 KSH at the actuarially fair price and 4,963 KSH at the market price for those offered the HBR insurance. This equates to a 50% increase in preferred coverage at the actuarially fair price and a greater than 100% increase at the market price. Among farmers offered LBR insurance, we see a negative but insignificant effect (-1,098 KSH) of playing the experiential game on preferred coverage levels at the actuarially fair price and a very large negative and statistically significant effect (-5,217 KSH) of playing the experiential game on preferred coverage levels at the market price.

Similar to the analysis at all prices, we find that on average farmers preferred 2,260 KSH

Table 3. Treatment Effects on Preferred Coverage Level

	All prices	Actuarially fair	Market price
No game, high basis risk	13,460.2 (706.4)	10,963.1 (1,125.3)	3,954.9 (615.8)
Average game effect	1,291.9 (-1085.2, 3669.0)	2,229.2 (-816.3, 5274.8)	17.2 (-2533.0, 2567.4)
Game effect with high basis risk insurance	3,863.6** (480.4, 7246.8)	5,373.7*** (1358.5, 9388.8)	4,962.6*** (1318.9, 8606.2)
Game effect with low basis risk insurance	-1,429.8 (-4732.5, 1873.0)	-1,098.7 (-5759.8, 3562.5)	-5,216.7*** (-8953.7, -1479.6)
Average effect of low basis risk insurance	2,292.2 (-524.1, 5216.7)	2,260.3 (-1106.3, 5848.8)	2,219.6* (-276.2, 5173.5)
Low basis risk insurance effect, no game	4,961.1*** (1755.7, 8749.3)	5,523.7*** (1927.7, 9563.3)	7,352.0*** (2814.2, 12027.3)
Low basis risk insurance effect, with game	-332.2 (-4165.4, 3662.9)	-948.7 (-6760.9, 5358.0)	-2,827.2** (-5551.2, -105.8)
Observations	3,409	487	487

Note. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. 95% confidence intervals in brackets. Confidence intervals for the experiential game effects are based on robust standard errors clustered by farmer. Confidence intervals for the effects of low basis risk insurance are based on the wild cluster bootstrap with two-point Rademacher weights (Roodman et al. 2019). Covariates include age, education, literacy, gender, land size, number of droughts in the past five years, loan availability, formal savings, planted acreage, projected net value of production, crop (green gram or sorghum), and enumerator fixed effects. The first row gives sample averages for farmers who did not play the experiential game and were offered the high basis risk product, with the standard error of the mean in parentheses.

more coverage when LBR insurance is offered than HBR insurance at the market price and 2,220 KSH more coverage at the market price, although the difference is only significant at the market price. When we consider only farmers who did not play the experimental game, we find much larger differences of 5,524 KSH at the actuarially fair price and 7,532 KSH at the market price, which amount to approximately a 50 and 200% increase. When we only consider farmers who did play the experimental game, we see a small but insignificant difference (−949 KSH) at the actuarially fair price and a negative and significant difference at the market price (−2,827 KSH). Note that in part A of the online supplementary appendix we check the robustness of our results for impacts on knowledge and preferred coverage to sample construction, regression specification, and mode of inference. None of our conclusions are altered by changing our empirical approach.

As discussed above, we expected the experiential game to be more effective at increasing insurance coverage for LBR insurance than for HBR insurance, and similarly we expected the wedge in preferred coverage between LBR and HBR insurance to be greater for those who played the experiential game than for those who did not. We discuss possible reasons for these surprising findings in the next section.

Exploration of Basis Risk and Experiential Game Interaction Effects

Our empirical results demonstrate that the experiential game modestly increased knowledge and that farmers can distinguish between LBR and HBR insurance. The latter finding reveals sensitivity of preferred coverage levels to basis risk. Even though index insurance is a complicated contract, we show smallholder farmers with limited education are sophisticated enough to discern important terms of the contract and adjust their valuation of the product accordingly. These results were not *ex ante* obvious, and as such, each of these findings make an important contribution to the literature on agricultural insurance demand.

On the other hand, the estimated impacts of the experiential game conditional on insurance type and LBR insurance conditional on game type are puzzling. As described above, the experiential game ought to increase the salience of basis risk and could therefore decrease preferred coverage levels conditional

on insurance type. But given that the experiential game succeeded to a modest degree in teaching farmers about basis risk, we expected the experiential game to have a greater positive effect on preferred coverage levels for LBR insurance, a greater negative effect for HBR insurance, or both. Instead, the experiential game effect is positive among farmers assigned to HBR insurance and either indistinguishable from zero or negative for farmers assigned to LBR insurance, as shown in rows 3 and 4 of table 3. The impact of being offered LBR instead of HBR insurance conditional on game type is just as surprising, given that we expected to see minimal impact of insurance type on farmers playing the placebo game and a positive effect of LBR insurance among farmers playing the experiential game. Instead, the sixth and seventh rows of table 3 show large positive effects of LBR insurance on preferred coverage levels among farmers who played the placebo game and effects that are either indistinguishable from zero or negative among farmers playing the experiential game. Below we consider several explanations for our surprising results.

We begin by ruling out two potential explanations. First, our results would make sense if farmers playing the experiential game with LBR insurance happened to experience more negative basis risk events or fewer positive basis risk events than farmers playing the experiential game with HBR insurance by chance. But as already discussed, being assigned to LBR reduced the number of negative basis risk events during the experiential game and had no effect on the number of positive basis risk events, and these results hold regardless of whether we consider all rounds or only those rounds in which a given farmer purchased insurance.

Another explanation that we can rule out is that the LBR maps (figure 2a) were less effective than the HBR maps (figure 2b) at conveying information. The former are visually busier than the latter and might be more difficult to learn from. However, we received no indication that this was the case during pretesting or implementation. As a check, we test whether the HBR version of the experiential game was more effective at improving knowledge than the LBR version. We do this by estimating equation (2) with the knowledge index and answering each knowledge question correctly as dependent variables (table A5 in the online supplementary appendix). We do not find the LBR experiential game to be any less

effective than the HBR version. In fact, we see that the increase in the knowledge score from playing the experiential game is concentrated among farmers offered LBR insurance, a result of the fact that the HBR version of the experiential game had a significant negative effect on the probability of correctly answering that index insurance “may pay out even with enough rain” (i.e. positive basis risk events may occur). For all other knowledge questions, point estimates are similar for both insurance types among farmers playing the experiential game. Overall, these results do not suggest learning outcomes were better with the HBR version of the experiential game.

Next we turn to two explanations that we are unable to rule out. Farmers may have a strict budget constraint on insurance, and being assigned to LBR insurance or the experiential game could be sufficient to reach this constraint. Under this scenario, there is no additional impact of playing the experiential game when LBR insurance is offered and no additional impact of being offered LBR insurance for those playing the experiential game. This argument is weakened somewhat by the fact that the combination of treatments (LBR and the experiential game) appears to, if anything, reduce preferred coverage levels if relative to each treatment in isolation, at least at high prices, and the fact that few farmers ultimately bought insurance.

The final possibility we consider is that among experiential game participants, HBR insurance exceeded expectations while LBR insurance did the opposite, influencing farmers’ valuations of the two products. The large average treatment effect of offering LBR insurance to farmers who only received basic information and played the placebo game (see table 3) suggests that the gap in the perceived quality of the two products was quite large before starting the experiential game, perhaps setting the stage for LBR insurance to disappoint experiential game participants, whereas HBR insurance performed better than expected. Gagnon-Bartsch and Bushong (2019) build on earlier work by Tversky and Kahneman (1979); Bell (1985); and Koszegi and Rabin (2006) to develop a learning model where an agent misattributes feelings of elation or disappointment stemming from surprisingly good or bad information about something to the value of the thing itself (e.g. the quality of a good, the effectiveness of a technology, the difficulty of a task, etc.), leading to

misattribution bias. Bushong and Gagnon-Bartsch (2019) find empirical evidence in support of their model using a Mechanical Turk experiment where subjects who are randomly assigned to an easy task are more willing to repeat it than subjects who knew they would face the same easy task with certainty, whereas subjects who are randomly assigned to an onerous task were less willing to repeat it than subjects who knew they would face the same onerous task with certainty. We apply the Gagnon-Bartsch and Bushong (2019) model to the process of learning about insurance performance through an experiential game. The model illustrates how farmers could exhibit equal or higher demand for an inferior, but pleasantly surprising, HBR insurance product compared to a superior, but disappointing, LBR product.

Suppose that after the basic information session, a farmer’s expectations about the quality of insurance product $c \in \{HBR, LBR\}$ are given by θ_c . In this way, θ_c acts as a reference point when processing new information about the farmer’s assigned product. By playing the experiential game, the farmer experiences a series of weather and insurance outcomes that can be summarized by the scalar measure of insurance quality x_c . Using the gain–loss utility framework developed by Koszegi and Rabin (2006), which builds from the value function of Tversky and Kahneman (1979), the farmer’s utility from the outcome of the experiential game, u , can be written as:

$$(3) \quad \begin{aligned} u(x_c | \theta_c) &= x_c + \eta n(x_c | \theta_c) \\ \text{where} \\ n(x_c | \theta_c) &= \begin{cases} (x_c - \theta_c) & \text{if } x_c > \theta_c \\ \lambda(x_c - \theta_c) & \text{if } x_c < \theta_c \end{cases} \end{aligned}$$

That is, the farmer’s utility from the insurance product in the experiential game is a linear combination of insurance quality during the experiential game and a gain–loss utility term, $n(x_c | \theta_c)$, where the latter is weighted by $\eta > 0$. We assume that utility is linear in x_c for simplicity, but nothing would change if we instead used a nonlinear transformation of x_c . If $\lambda > 1$ then the participant is loss averse, that is, losses loom larger than gains in the utility function. Individuals assigned to the placebo game designed to elicit prospect theory parameters have a mean λ of 3.99, with two-thirds of the sample exhibiting loss aversion (Shin et al. 2020).

After completing the experiential game, farmers update their beliefs about insurance quality, and then give their preferred coverage values at each price in the auction. A rational farmer would ignore gain–loss utility in updating her beliefs, because gains and losses relative to a reference point have nothing to do with the intrinsic qualities of insurance (e.g. its risk reduction potential or expected payout). We assume that rational farmers follow Bayes rule so that updated beliefs are a convex combination of initial beliefs about quality and perceived quality as experienced in the experiential game:

$$(4) \quad \theta_c^{post} = \alpha x_c + (1 - \alpha) \theta_c$$

where $\alpha < 1$ and θ_c^{post} represents updated beliefs about insurance product c .¹¹

Subtracting θ_{HBR}^{post} from θ_{LBR}^{post} and rearranging yields the following condition that has to hold for the farmer to prefer HBR insurance to LBR insurance after updating her beliefs:

$$(5) \quad (x_{LBR} - x_{HBR}) + \frac{1 - \alpha}{\alpha} (\theta_{LBR} - \theta_{HBR}) < 0$$

Because LBR is a higher quality product, it is reasonable to assume that LBR insurance will perform better in the experiential game, so that $x_{LBR} > x_{HBR}$. Similarly, the estimated effects of offering LBR insurance to farmers who only receive basic information suggest that $\theta_{LBR} > \theta_{HBR}$. Under these two assumptions, both terms on the left hand side of equation 5 are positive, and the inequality therefore cannot hold. A rational farmer will never prefer LBR insurance to HBR insurance if the former is perceived to be superior before the experiential game and also performs better in the experiential.

Now suppose that instead of updating rationally, the farmer incorrectly remembers how her experience in the experiential game was influenced by reference dependence. Although the farmer is aware that utility from the experiential game is a combination of two

terms, she incorrectly weights $n(x_c | \theta_x)$ by $\hat{\eta} \in [0, \eta)$ instead of η when attempting to back out x_c from $u(x_c | \theta_x)$. As initially proposed by Gagnon-Bartsch and Bushong (2019), incorrect weighting of gain–loss utility leads the farmer to mistakenly recall insurance quality in the experiential game as:

$$\hat{x}_c = \begin{cases} x_c + \kappa^G (x_c - \theta_c) & \text{if } x_c > \theta_c \\ x_c + \kappa^B (x_c - \theta_c) & \text{if } x_c < \theta_c \end{cases}$$

(6) where $\kappa^G = \frac{\eta - \hat{\eta}}{1 + \hat{\eta}}$ and $\kappa^B = \lambda \frac{\eta - \hat{\eta}}{1 + \lambda \hat{\eta}}$

That is, the farmer misattributes a portion of gain–loss utility to the quality of her assigned insurance product as experienced in the experiential game. In equation 6, κ^G and κ^B are misattribution weights for good and bad surprises, respectively. Both are increasing in the weight placed on gain–loss utility (η) and in misattribution severity (i.e. decreasing in $\hat{\eta}$), while κ^B is increasing in loss aversion (λ). For loss averse farmers, $\kappa^B > \kappa^G$ so that negative surprises lead to greater misattribution.

For a given insurance product, denote $\kappa_c \equiv \kappa^G 1\{x_c > \theta_c\} + \kappa^B 1\{x_c < \theta_c\}$. Replacing x_c with $\hat{x}_c$ in equation 4, we find that with misattribution a farmer will update her beliefs as follows:

$$(7) \quad \hat{\theta}_c^{post} = \alpha(1 + \kappa_c)x_c + (1 - \alpha(1 + \kappa_c))\theta_c$$

A comparison of $\hat{\theta}_{HBR}^{post}$ and $\hat{\theta}_{LBR}^{post}$ shows that a misattributing farmer will prefer the HBR product to LBR insurance if the following inequality holds:

$$(8) \quad (x_{LBR} - x_{HBR}) + \frac{(1 - \alpha)}{\alpha} (\theta_{LBR} - \theta_{HBR}) < \kappa_{HBR}(x_{HBR} - \theta_{HBR}) - \kappa_{LBR}(x_{LBR} - \theta_{LBR})$$

The left hand side of equation 8 is identical to the left hand side of equation 5. But instead of having zero on the right hand side of the inequality in equation 8, we have a new term arising from misattribution that distorts the comparison of the two insurance products. Whereas the inequality in equation 5 cannot hold, equation 8 can hold if HBR insurance exceeds expectations ($x_{HBR} > \theta_{HBR}$), LBR insurance falls short of expectations ($x_{LBR} < \theta_{LBR}$), or both, implying that HBR insurance can be preferred to LBR insurance under misattribution.

¹¹See Chambers and Healy (2012) for conditions sufficient to ensure that a Bayesian learner's updated beliefs are a convex combination of initial beliefs and the new information, as we have assumed here. Also, we assume that α is the same for both insurance types. This would follow if (for example) θ_c and x_c were independent, normally distributed random variables whose variances do not depend on insurance type. Then for insurance type c the weight placed on x_c when updating would be equal to the variance of θ_c divided by the sum of the variances of θ_c and x_c .

Whether the inequality in equation 8 holds will depend on the gap between initial beliefs and performance in the experiential game for the two insurance types, as well as α , η , $\hat{\eta}$, and λ . Increasing α so that more weight is placed on experiential game outcomes when updating beliefs brings the second term on the left hand side of equation 8 toward zero, making it more plausible that HBR insurance is preferred to LBR insurance. If LBR insurance falls short of expectations while HBR exceeds them, then $\kappa_{LBR} = \kappa^B$ and $\kappa_{HBR} = \kappa^G$. Under this scenario, the right hand side of equation 8 is increasing in η , decreasing in $\hat{\eta}$, and increasing in λ . In other words, greater weight on gain-loss utility, larger misattribution errors, and higher loss aversion will all make the inequality in equation 8 more likely to hold when LBR insurance is underwhelming and HBR insurance outperforms expectations.

Although we cannot formally test whether the model above explains the surprising aspects of our results, the large treatment effects of offering LBR insurance on preferred coverage levels among farmers who did not play the experiential game suggest that θ_{LBR} is much greater than θ_{HBR} , and therefore the model provides a plausible explanation for our findings. Future experimental research could attempt to capture priors on insurance quality as well respondent perception of gain-loss utility in an attempt to rigorously test the role of misattribution bias in insurance demand.

Conclusion

Index insurance, including weather index insurance (Hazell 1992) and area yield insurance (Miranda 1991), has been promoted as a viable low cost alternative to standard indemnity crop insurance. In the U.S. a newly proposed weather index-based crop disaster program is estimated to save taxpayers \$3–4 billion per year relative to the current crop insurance program, a reduction of approximately 30% (Belasco et al. 2020). Although index insurance can be cheaper, basis risk—the mismatch between index insurance payouts and individual outcomes—means a payout is not guaranteed in the event of an individual loss. Demand for index insurance is extremely low in most instances, with some evidence suggesting basis risk is a major barrier to index insurance uptake. Furthermore,

insurance is a complex financial instrument and complete understanding of its features can be difficult to achieve. Farmers may not be willing to purchase an insurance product that they do not understand.

We conducted a randomized control trial with 487 smallholder farmers in Kenya to test farmers' sensitivity to basis risk and explore whether an experiential game can teach farmers about index insurance and raise preferred coverage levels for insurance. During the study period, farmers attended morning or afternoon sessions. Each session was randomly assigned to carry out activities with either a high basis risk or low basis risk insurance product. All session participants attended a basic information presentation about index insurance. Following the basic information session, half of each session group was randomly selected to participate in an experiential insurance game simulating their assigned insurance product and highlighting the role of basis risk. The other half of each session group was assigned to a placebo game. To measure preferred insurance coverage levels, we used an experimental auction for weather index insurance corresponding to the assigned insurance product at the conclusion of the games.

We find that playing the experiential game increased the number of index insurance questions answered correctly by 4.4% on a four-question quiz, even though understanding was already high after the basic information session alone. This finding, on its own, is of crucial importance to policymakers and practitioners. As Platteau et al. (2017) point out in a recent review, "A huge challenge for microinsurance schemes is that they must be properly understood by potential and actual clients. The core concept of insurance, spending money in return for an uncertain payout covering a hypothetical event is, indeed, not straightforward."

Although our study and that of Cai and Song (2017) suggest that experiential games show promise in teaching farmers about insurance and may increase preferred coverage, there is still much to be learned. One question of interest is whether materials from the experiential game (the maps) are useful for explaining and promoting weather index insurance without the actual game play, which is costly in terms of time and money. Cai and Song (2017) find that experiential games were no more effective at promoting insurance demand than carefully explaining drought

probabilities to farmers. Whether experiential learning is key to turning farmers into sophisticated purchasers of weather index insurance, or if key aspects of weather index insurance need only be packaged in a comprehensible way, is a rich area for future research.

We also shed light on the question of whether farmers are able to differentiate between a high and low quality product. Using the auction data, we find that farmers are highly sensitive to basis risk. Among farmers who did not play the experiential game, preferred insurance coverage levels for low basis risk insurance is 37% higher than for high basis risk insurance. For farmers offered high basis risk insurance, playing the experiential game increases preferred insurance coverage by 29%. This finding is highly policy relevant, because the available evidence suggests basis risk is a prominent feature of existing index insurance products in both developed and developing economies (Clarke 2016; Jensen et al. 2016; Yu et al. 2019).

The interaction effect between LBR insurance and the experiential game is negative and significant, with the net result being that the two treatments are no more effective at increasing preferred coverage levels when combined than they are individually. We conclude index insurance product quality and knowledge may be substitutes rather than complements with respect to increasing preferred coverage levels. We draw on economic theory and provide some discussion for why this may be, but we leave further empirical work to a future application.

Despite the fact that our treatments were effective in isolation at increasing preferred coverage in an experimental auction, actual uptake of insurance at the close of the study was near zero. Thus we cannot say that either the offer of lower basis risk insurance or the use of an experiential game increased insurance purchases. This study thus contributes to an increasingly large literature indicating farmers will not purchase weather index insurance as a standalone product. Recent exceptions to the rule of low demand for weather index insurance include Vasilaky et al. (2019), Casaburi and Willis (2018), and Belissa et al. (2019). But each of these exceptions is characterized by unique circumstances. In Vasilaky et al. (2019), farmers were first given money that they could use to purchase insurance before being asked to purchase it themselves the subsequent season. In Casaburi and Willis (2018), contract farming made it

possible for the insurer to finance the premium and then deduct it from payouts to farmers at the end of the season. Belissa et al. (2019) find that delayed premium payment without any contract farming agreement increases demand substantially, although default rates are relatively high putting into question the financial viability of this approach. Farmers in our study were involved in contract farming, but we did not offer insurance through their contract farming arrangements, and they were required to pay premiums before planting and not at harvest.

Uninsured risk remains a barrier to investment and production globally, and climate change threatens to increase the global agriculture system's exposure to weather shocks. Will weather index insurance be a broadly applicable risk management solution or a niche product for specific circumstances? Only time will tell. But if index insurance is to ever become a widely adopted tool, our study highlights the importance of reducing basis risk and illustrates how experiential games can be used to improve product understanding.

Supplementary Material

Supplementary material are available at *American Journal of Agricultural Economics* online.

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